

KCPC Week 2 Solutions

1 Wishful Thinking

Basic – Complete The Square

Subtract the right-hand side from the left:

$$x^2 + y^2 - \frac{(x+y)^2}{2} = \frac{(x-y)^2}{2} \geq 0.$$

Equality holds exactly when $x = y$.

Key insight: Wish for a square because squares are automatically non-negative.

Easy – Make The Sum Telescope

Use partial fractions:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Therefore all intermediate terms cancel:

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

Key insight: Rewriting each term as a difference exposes cancellation.

Medium – A Number Puzzle

Set $x = u + v$ and $y = u - v$. The equation becomes $u^2 + v^2 = z^2$. Take the Pythagorean triple

$$u = m^2 - 1, \quad v = 2m, \quad z = m^2 + 1$$

for any integer $m \geq 3$. Then

$$x = m^2 + 2m - 1, \quad y = m^2 - 2m - 1, \quad z = m^2 + 1$$

is positive and satisfies the original equation. Varying m gives infinitely many solutions.

Key insight: The substitution turns the desired equation into a familiar one.

Hard – Improve The Recurrence

Define $b_n = a_n + 1$. Then

$$b_{n+1} = a_{n+1} + 1 = 2a_n + 2 = 2b_n, \quad b_1 = 2.$$

Hence $b_n = 2^n$, so

$$a_n = 2^n - 1.$$

Key insight: Changing variables absorbs the awkward constant term.

Very Hard – Factor The Equation

Multiplying by nxy gives $n(x + y) = xy$. Rearranging and adding n^2 to both sides gives

$$(x - n)(y - n) = n^2.$$

Both x and y must exceed n . Thus, for every positive divisor d of n^2 , set

$$x = n + d, \quad y = n + \frac{n^2}{d}.$$

These are all ordered positive solutions, since every factorisation of n^2 arises this way.

Key insight: Adding the right term changes a sum equation into a factorisation.

2 Symmetry

Basic – Chomp: Poisoned Chocolate

The first player eats the single square at (m, n) (top-right corner), leaving a symmetric L-shape. Thereafter, the first player mirrors the opponent's move by 180° rotation about the centre of the remaining L-shape. The mirror is always legal and avoids the poisoned corner. The first player never eats the poisoned square first, so wins.

Key insight: Create a symmetric position, then mirror the opponent's moves.

Easy – Coins On A Circular Table

The first player places a coin at the centre. After every later move, the first player places a coin in the position obtained by rotating the opponent's coin by 180° . The table remains symmetric after each response, so the mirrored placement is legal whenever the opponent's placement is legal. The first player therefore moves last.

Key insight: Occupy the unique fixed point, then pair all remaining positions.

Medium – A Symmetric Integral

Let $I = \int_0^1 f(x) dx$. Substituting $u = 1 - x$ gives

$$I = \int_0^1 f(1 - x) dx.$$

Adding the two expressions and using the given identity,

$$2I = \int_0^1 (f(x) + f(1 - x)) dx = \int_0^1 1 dx = 1.$$

Hence $I = \frac{1}{2}$.

Key insight: Reflecting the interval pairs every value with its complement.

Hard – Odd Skew-Symmetric Matrices

Using invariance of the determinant under transposition,

$$\det A = \det A^T = \det(-A) = (-1)^n \det A.$$

When n is odd this says $\det A = -\det A$, so $2 \det A = 0$ and $\det A = 0$.

Key insight: Transpose symmetry turns the determinant into its own negative.

Very Hard – Six People

Represent people by vertices and colour each edge red for acquaintances and blue for strangers. Choose a vertex v . At least three of its five incident edges have one colour; by symmetry suppose va, vb, vc are red. If one of ab, bc, ca is red, it forms a red triangle with v . Otherwise ab, bc, ca are all blue and form a blue triangle.

Key insight: Colour symmetry reduces two relationship cases to one.

3 Extreme Principle

Basic – Erdős–Szekeres Monotone Subsequence

For each term a_i , let L_i be the length of the longest increasing subsequence starting at a_i , and D_i the longest decreasing. If $a_i < a_j$ with $i < j$, then $L_i > L_j$; if $a_i > a_j$, then $D_i > D_j$. Thus all pairs (L_i, D_i) are distinct. Each L_i, D_i is between 1 and n (otherwise we have the desired subsequence). There are at most n^2 distinct pairs, but $n^2 + 1$ terms. By pigeonhole, some pair repeats, contradiction unless a subsequence of length $n + 1$ exists.

Key insight: Map each term to a pair of lengths; pigeonhole principle forces a collision unless the bound is reached.

Easy – Minimum Degree Forces A Cycle

Choose a longest path $v_0v_1 \dots v_k$. Every neighbour of v_k must already lie on the path, since a new neighbour would extend it. Because v_k has at least two neighbours, it is adjacent to some v_i with $i < k - 1$. The edge v_kv_i together with the path from v_i to v_k forms a cycle.

Key insight: An endpoint of a longest path cannot escape to a new vertex.

Medium – An Empty Triangle

Among all non-degenerate triangles whose vertices belong to the set, choose one of minimum positive area. If another given point lay in its interior, joining that point to the three vertices would split the triangle into three smaller triangles of positive area. At least one would contradict the chosen minimum. Therefore its interior is empty.

Key insight: A minimum-area counterexample cannot contain a smaller one.

Hard – A Path Through A Tournament

Choose a longest directed path $v_1 \rightarrow \dots \rightarrow v_k$. Suppose a vertex u is omitted. If $u \rightarrow v_1$, prepend u ; if $v_k \rightarrow u$, append u . Otherwise $v_1 \rightarrow u$ and $u \rightarrow v_k$. Moving along the path, there is an index i with $v_i \rightarrow u$ and $u \rightarrow v_{i+1}$, so u can be inserted between them. Every case contradicts maximality, so the path contains every vertex.

Key insight: A longest object is rigid enough that an omitted vertex causes a contradiction.

Very Hard – Longest Paths Meet

Let two longest paths P and Q both have L edges. Suppose they are disjoint. Since the graph is connected, choose a shortest path joining a vertex p of P to a vertex q of Q , with no other vertices on $P \cup Q$. From p , one endpoint of P is at distance at least $L/2$ along P ; similarly, one endpoint of Q is at distance at least $L/2$ from q . Joining these two subpaths through the connecting path produces a path longer than L , a contradiction.

Key insight: Two disjoint extremes can be joined to create something more extreme.

4 Reduction

Basic – An Acquaintance Graph

Represent each person by a vertex and join two vertices when those people are acquainted. Every degree is one of $0, 1, \dots, n - 1$. However, degrees 0 and $n - 1$ cannot both occur: if one person knows everyone, nobody knows zero people. Thus at most $n - 1$ different degree values occur among n vertices. By the pigeonhole principle, two vertices have the same degree.

Key insight: The graph representation turns acquaintances into vertex degrees.

Easy – The Bridges Of Königsberg

Replace each land mass by a vertex and each bridge by an edge. A walk using every bridge exactly once would be an Euler trail. In such a trail, every vertex except possibly the start and finish has even degree. The Königsberg graph has four odd-degree vertices, so no Euler trail exists.

Key insight: Only incidence matters, so geography can be discarded.

Medium – Missionaries And Cannibals

State: (M_L, C_L, B) with M missionaries, C cannibals on left, $B \in \{L, R\}$ boat position. Safe iff $M = 0$ or $M \geq C$ on each bank. Graph edges: valid boat moves (1-2 people). BFS from $(3, 3, L)$ to $(0, 0, R)$ yields: MM cross, M returns, CC cross, C returns, MM cross, C returns, CC cross, M returns, MM cross. All intermediate states safe.

Key insight: State-space graph with safety constraints; BFS finds valid path.

Hard – Lamps As Linear Algebra

Encode on/off by 1/0 in $\mathbb{F}_2$. Let v be the initial state and let column i of an $n \times n$ matrix A have ones in positions $i - 1, i, i + 1$ modulo n . If x_i records whether move i is used, then reaching all off is equivalent to

$$Ax = v \quad \text{over } \mathbb{F}_2.$$

Gaussian elimination decides whether this system is consistent and, when it is, returns a sequence of moves. Repeating a move is unnecessary because doing it twice has no effect.

Key insight: Toggle operations become vector addition modulo 2.

Very Hard – Assignment As Maximum Flow

Create a source s , one vertex for each student, one for each project, and a sink t . Add capacity-1 edges from s to every student, from a student to each acceptable project, and from every project to t . An integral flow of value n selects one outgoing student–project edge for every student and uses every project at most once. Conversely, any valid assignment gives such a flow. Therefore an assignment exists exactly when the maximum flow has value n .

Key insight: Capacity constraints encode distinct choices, reducing assignment to flow.

Advanced Material

The remaining solutions cover optional extensions. Their theory and worked examples are presented in *Week 2 Advanced Slides*.

5 Advanced Logic And Common Knowledge

Basic – Blue-Eyed Islanders ($k = 2$)

If $k = 1$, the single blue-eyed islander sees no blue eyes, knows the announcement refers to them, and leaves night 1. If $k = 2$, each blue-eyed islander sees one blue-eyed person. If I had brown eyes, the other would see 0 blue eyes and leave night 1. They don't. So I must have blue eyes. Both leave night 2.

Key insight: Public announcement creates common knowledge; induction on k gives the day of departure.

Easy – Cheating Husbands

Let k be the number of cheating husbands. By the same induction as Blue-Eyed Islanders, all k wives deduce their husbands cheated on day k and act at midnight. If $k = 0$, the queen's announcement is false, but the logic still holds: no one acts because the premise is violated.

Medium – Sum And Product Puzzle

P doesn't know $\rightarrow$ product is not a product of two primes (or 4). S knew $\rightarrow$ sum cannot be written as sum of two primes (Goldbach). Sums like 11, 17, 23, 27, 29, 35, 37, 41, 47, 53 work. P now knows $\rightarrow$ among factor pairs of xy , only one has a sum from the allowed list. S now knows $\rightarrow$ among pairs summing to $x + y$, only one has a product with the above property. The unique solution: $x = 4$, $y = 13$ (sum 17, product 52).

Key insight: Iterated elimination using meta-knowledge; each statement restricts the set of possible pairs.

Hard – Cheryl's Birthday

Dates: May 15, 16, 19; June 17, 18; July 14, 16; Aug 14, 15, 17. Albert knows month, Bernard knows day. Albert: Bernard doesn't know $\rightarrow$ month is not May or June (19 and 18 are unique days). Remaining: July 14, 16; Aug 14, 15, 17. Bernard now knows $\rightarrow$ day is not 14. Remaining: July 16; Aug 15, 17. Albert now knows $\rightarrow$ month must be July (if August, Albert couldn't distinguish 15 and 17). Answer: July 16.

Very Hard – 100 Prisoners And A Light Bulb

Designate one prisoner as the Counter. Others are Drones. Protocol: A Drone who has never turned the light on, and finds it off, turns it on exactly once. The Counter only turns the light off, and increments a mental count each time they find it on. When the Counter's count reaches 99, they announce. Proof: Each Drone eventually visits and

turns the light on if they haven't. The Counter eventually sees each on-state. The count reaches 99 exactly when all 99 Drones have visited.

Key insight: The light bulb encodes one bit of shared state; the Counter aggregates information.

6 Advanced Invariants And Monovariants

Basic – Peg Solitaire (Small Board)

On a 3×3 board with centre empty, label holes with values $1, x, x^2$ where $x^2 + x + 1 = 0$. Assign weights by coordinates mod 3. Initial sum $\neq$ final sum (centre hole), so impossible.

Key insight: Polynomial invariant over $\mathbb{C}$ captures the jump structure.

Easy – Token Game Modulo 3

Assign weight 2^{position} modulo 3. Initially: $2^0 + 2^1 = 1 + 2 = 0 \pmod{3}$. A move replaces tokens at x and $x + 1$ (total weight $2^x + 2^{x+1} = 3 \cdot 2^x \equiv 0$) with one at $x + 2$ (weight $2^{x+2} = 4 \cdot 2^x \equiv 2^x$). Wait – this doesn't work! Correct invariant: Assign weight 1 to position 0, x to position 1, x^2 to position 2, etc., with x a root of $x^2 + x + 1 = 0$. Then $x^k + x^{k+1} = x^k(1 + x) = -x^{k+2}$. The sum modulo the polynomial is invariant. Position 100 weight is $x^{100} = x^{100 \bmod 3} = x$. Initial sum $1 + x \neq x$, so impossible.

Medium – Conway's Soldiers

Assign weight $\varphi^{|y|}$ where $\varphi = (\sqrt{5} - 1)/2 \approx 0.618$ (so $\varphi^2 = 1 - \varphi$). A vertical jump: two tokens at y and $y + 1$ become one at $y + 2$. Weight change: $\varphi^y + \varphi^{y+1} - \varphi^{y+2} = \varphi^y(1 + \varphi - \varphi^2) = 0$. Horizontal jumps don't change y . Total weight never increases. Initial weight for $y \leq 0$ is finite; weight at $y = 5$ is φ^5 . Sum of all weights below line exceeds any single target weight, but the invariant shows you cannot concentrate enough weight at $y = 5$. Actually, the standard proof uses $x = (\sqrt{5} - 1)/2$ and shows total weight of starting configuration is 1, while weight needed at row 5 is $\varphi^5 \approx 0.09$, but the row 4 pegs needed to get there consume the weight. Full proof: each row's total weight is bounded by a geometric series.

Hard – Polynomial Invariant For (x+2,y-1)

The move $(x, y) \rightarrow (x + 2, y - 1)$ changes $x + 2y$ by $2 + 2(-1) = 0$. Thus $P(x, y) = x + 2y$ is a polynomial invariant. More generally, any polynomial in $x + 2y$ is invariant. The reachable positions are exactly those with the same value of $x + 2y$ as the start: $x + 2y = 0$ (and $x, y \geq 0$ reachable by non-negative steps).

Key insight: The linear combination $x + 2y$ is unchanged by the move; polynomial invariants characterise reachable sets.

Very Hard – IMO 1986 Problem 3 (Sequence Invariant)

Define $d_n = a_n^2 - a_{n-1}a_{n+1}$. Then $d_{n+1} = a_{n+1}^2 - a_na_{n+2}$. Using $a_{n+2} = a_{n+1} + a_n + 1$, compute $d_{n+1} - d_n$. After algebra, $d_{n+1} = d_n$. So invariant modulo 3. Base: $d_1 = a_1^2 - a_0a_2 = 1^2 - 0 \cdot a_2 = 1$. Thus $d_n \equiv 1 \pmod{3}$ for all n .

7 Colouring And Tiling: Generalisations

Basic – Mutilated Chessboard: Four Corners Removed

An 8×8 board with four corners removed has 60 squares. Colour the board with three colours in a repeating pattern: colour (i, j) by $(i+j) \bmod 3$. A domino covers one square of each of two colours (if horizontal) or the same colour twice (if vertical)? Actually, a horizontal domino covers (i, j) and $(i, j+1)$ with colours c and $c+1$, a vertical covers (i, j) and $(i+1, j)$ with colours c and $c+1$. Wait – a domino always covers two squares of different colours in this colouring. So each domino covers one square of each of two colours. Count colour frequencies: 20 squares of each colour (64 total, 4 corners = 1 of each colour removed, so 20 each). 30 dominoes would cover 30 squares of each colour – but we only have 20 of each. Impossible.

Key insight: Three-colouring reveals colour imbalance: dominoes cover two colours, but counts are unequal.

Easy – Tiling With T-Tetrominoes

Colour the board by $(x \bmod 2, y \bmod 2)$: four colours 00, 01, 10, 11. A T-tetromino covers 3 of one colour and 1 of another (or 2 and 2 depending on orientation). Count colour frequencies on 4×4 : 4 of each colour. Any T-tetromino tiling would require colour counts to be multiples of the pattern, which is impossible. For $4 \times n$, similar parity argument fails for some n ; the exact condition is that n must be a multiple of 4.

Medium – Fault-Free Domino Tiling

6×6 : Construct a spiral tiling with no vertical or horizontal fault line. 6×8 : Colour columns alternately black and white. A fault line would require each row to have an even number of horizontal dominoes crossing it. Counting parity shows contradiction. (Standard proof: consider the board as three 2×8 strips; each strip needs even horizontal dominoes, but total horizontal dominoes parity conflicts.)

Hard – Triominoes On A Chessboard

Colour the 8×8 board with 3 colours in a repeating pattern: colour (i, j) by $(i+j) \bmod 3$. A 1×3 tromino covers one of each colour. The board has 64 squares; removing one corner leaves 63 squares with colour counts 22, 21, 20 (not equal). Impossible.

Very Hard – Generalised Mutilated Chessboard

An $m \times n$ board with two opposite-colour squares removed can be tiled by dominoes iff mn is even and $(m, n) \neq (1, 2), (2, 1)$. Proof: If mn odd, total squares odd after removal,

impossible. For even mn , standard constructive proof: pair rows, snake through columns, route around holes. The only small obstructions are the trivial boards.

8 Induction And Recursion: From Proof To Algorithm

Basic – Josephus Problem

$J(1) = 1$. For $n > 1$, after first pass, even positions eliminated, problem reduces to $J(\lfloor n/2 \rfloor)$ with positions renumbered. Recurrence: $J(2k) = 2J(k) - 1$, $J(2k + 1) = 2J(k) + 1$. Closed form: write n in binary, rotate left by one bit. $J(n) = 2(n - 2^{\lfloor \log_2 n \rfloor}) + 1$.

Easy – Gray Codes

$G_1 = [0, 1]$. Given G_n , construct G_{n+1} by prefixing 0 to G_n and prefixing 1 to reverse of G_n . Proof by induction: consecutive strings in first half differ by induction; boundary between halves is $0G_n$ followed by $1\text{reverse}(G_n)$, differing only in first bit; second half consecutive differ by induction.

Medium – Recursive 15-Puzzle Parity

Given configuration, compute permutation parity of tiles (read row-major). Compute blank row from bottom. Solvable iff parity = blank row parity (for even-width boards). Algorithm: count inversions in $O(n^2)$ or $O(n \log n)$ with BIT.

Hard – Counterfeit Coin With n Weighings

Maximum coins: $(3^n - 3)/2$ for heavy/light unknown. Strategy: ternary search. Partition coins into three equal groups A, B, C . Weigh A vs B . If balance, counterfeit in C ; if tilt, counterfeit in heavier/lighter group and we know its type. Recurse. Each weighing gives 3 outcomes, so 3^n leaves. But we also need to distinguish heavy vs light, costing one bit per coin, giving $3^n/2$ bound. Precise bound: $(3^n - 3)/2$.

Very Hard – Hadamard Matrix Construction

$H_1 = [1]$. $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$. Proof: Rows of H_{2n} are either (r, r) or $(r, -r)$ for rows r of H_n . Orthogonality follows from orthogonality of H_n rows. By induction, H_{2^k} is Hadamard.

9 Algorithmic Bridges

Basic – A Maze As A Graph

Create one vertex for every open cell. Join two vertices when their cells share a side. Run BFS from the start, storing distance 0 there and assigning each newly discovered neighbour distance one greater than its parent. The finish distance is the minimum number of moves; if it is never discovered, no route exists. BFS is optimal because it

visits vertices in non-decreasing distance from the start, so the first path reaching any cell uses the fewest possible edges.

Easy – Greedy Activity Selection

Sort by finish time $f_1 \leq f_2 \leq \dots \leq f_n$. Pick $i_1 = 1$. For $k > 1$, pick first i_k with $s_{i_k} \geq f_{i_{k-1}}$. Proof by exchange: let O be an optimal solution with earliest first finish. If O doesn't pick the greedy first choice, exchange it; the new solution is at least as good and has earlier first finish. Repeat inductively.

Medium – DP From Induction: Coin Change

$f(s)$ = min coins to make sum s . Recurrence: $f(s) = 1 + \min_{c_i \leq s} f(s - c_i)$, with $f(0) = 0$, $f(s < 0) = \infty$. Tabulation: for $s = 1$ to n , for each coin $c_i \leq s$, update $f(s) = \min(f(s), 1 + f(s - c_i))$. Time $O(kn)$, space $O(n)$.

Hard – Sliding Puzzle State-Space Search

State encoding: permutation of tiles 1–5 plus blank position (6 locations). 720 permutations $\times 6$ blank positions = 4320 , but parity invariant halves it to 2160 reachable from any state. Actually 3×2 puzzle has $6!/2 = 360$ reachable states. BFS from goal state precomputes distance to goal for all states. Online: encode start state, look up distance, reconstruct path.

Very Hard – Greedy Scheduling With Deadlines

Sort jobs by decreasing profit. Initialise disjoint-set union (DSU) over time slots $1 \dots n$, where each slot points to itself; slot 0 points to 0 (sentinel). For each job in sorted order: find latest available slot $t = \text{find}(d_i)$. If $t > 0$, schedule job at t , union t with $t - 1$ (mark t occupied, next available is $t - 1$). If $t = 0$, skip job. Proof: exchange argument. If optimal solution doesn't include this job, either it has a slot free before deadline (exchange in), or all slots before deadline are filled with lower-profit jobs (exchange one out).